MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED JAU-1A/B

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 This specification covers the JAU-1A/B cartridge actuated initiator. (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-23615	Design and Evaluation of Cartridge Actuated Devices
MIL-C-81968	Cartridge, Impulse, CCU-23/B

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL 298AS116	Closed Chamber Assembly
DL 560AS100	Initiator, Cartridge Actuated JAU-1A/B
DL 652AS100	Cartridge, Impulse, CCU-23/B
275AS107	Initiator, Cartridge Actuated, JAU-1/B and JAU-1A/B, Packaging Assembly
275AS108	Container, Inner, Assembly

PUBLICATIONS

Bureau of Naval Weapons

WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations

49 CFR 171-178	Hazardous Materials Regulations, Department of Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 37 JAU-1A/B cartridge actuated initiators shall be manufactured using the same methods and procedures to be

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used for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.2 Conformance to documents. The JAU-1A/B cartridge actuated interior covered by this specification shall be manufactured in accordance with all drawings listed on DL 560AS100.

3.2.1 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referenced on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of initiator manufacture are specified on the drawings, the bidder's selection shall be clearly stated in the proposal.

3.3 Primary components. For the purpose of this specification the CCU-23/B impulse cartridge shall be considered a primary component. See 4.3.3.

3.3.1 Cartridge. The cartridges shall be furnished from lots which were manufactured within the 12-month period immediately preceding the assembly of the cartridges into the initiators. The cartridges shall be manufactured in accordance with the drawings listed on DL 652AS100 and shall be selected from lots which were accepted in accordance with MIL-C-81968 (AS). See 4.6.1.

3.4 Visual inspection. Initiators shall be free of the following visible defects: burrs, dents, deep scratches, cracks, illegible markings, corrosion, foreign matter, chipping or flaking of the protective finish, incomplete or damaged threads, and other defects which could adversely affect installation of ballistic performance. See 4.6.2.

3.5 Bridge circuit resistance. The bridge circuit resistance of each completely assembled initiator, when measured electrically as specified in 4.6.3, shall be not less than 1.0 ohm nor more than 2.0 ohms.

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3.6 Outer sleeve seating torque verification. A seating torque of 300 ± 10 inch pounds, applied to the outer sleeve of the assembled initiator, shall not cause movement between the inner and outer sleeves.

3.7 Ballistic requirements. Initiators shall meet the requirements listed below.

3.7.1 Pressure. The upper and lower limits of pressure in pounds per square inch shall be 1700 and 560, respectively.

3.7.2 Ignition time. The time between the application of firing current to the initiator and the start of pressure rise shall not exceed 0.050 second.

3.7.3 Misfire. The initiator shall fire when subjected to a direct current of 5 + 1.0 - 0.0 amperes.

3.7.4 Mechanical failure. All initiator components shall withstand firing with no mechanical failure which results in failure to contain the propellant gases. See 4.7.3.1.

3.8 Locked shut requirements. The initiator shall meet the requirements of 3.7.3 and 3.7.4 when fired under locked shut conditions.

3.9 Workmanship. Initiators shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes, and assembly operations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 37 JAU-1A/B cartridge actuated initiators shall be prepared. Of these initiators, 32 shall be expended in the tests listed below and five shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.2 through 3.6 of this specification. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-23615: 15 "g" shock (4), vibration (4), and 40-foot drop (6). The numerals in parentheses denote the number of initiators allocated for each test. These initiators (with the exception of the 40-foot drop initiators), after subjection to the specified treatments, shall be conditioned for a minimum of 6 hours at $70^{\circ} \pm 5^{\circ}$ F and fired in the fixture immediately after removal from conditioning. In addition, 18 initiators shall be conditioned for a minimum of 6 hours, 9 at $-65^{\circ} \pm 5^{\circ}$ F and 9 at $160^{\circ} \pm 5^{\circ}$ F, and fired immediately after removal from conditioning. Eight of these firings, 4 from -65° F conditioning and 4 from 160° F conditioning, shall be conducted in the test fixture. The 10 remaining initiators, 5 from -65° F conditioning and 5 from 160° F conditioning, shall be fired in the locked-shut condition as specified in 4.7.3. All test fixture firings shall be conducted in the test fixture of 4.7.2 and shall meet the requirements of 3.7.1 through 3.7.4. The locked-shut firings shall meet the requirements of 3.7.3 and 3.7.4. The firing current used in all firings shall be as specified in 4.7.2. The failure of any initiator to meet the acceptance requirements as stated above shall cause rejection of the preproduction sample.

4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 3,201 and 10,000 initiators, respectively. Only primary components from a single lot shall be used in a production lot of initiators; however, one primary component production lot may be used in more than one initiator production lot.

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4.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major or minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified as minor.

4.5 Quality conformance. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL 560AS100 and verification of the following requirements and tests:

Item	Nature of requirement verification	Applicable paragraphs	Classification
Primary component	Single lot	3.3, 4.3.3	(M101)
Cartridge	Age and acceptance	3.3.1, 4.6.1	(M102)
Initiator	Visual inspection	3.4, 4.6.2	(M103)
Initiator	Bridge circuit resistance	3.5, 4.6.3	(M104)
Initiator	Ballistic requirements	3.7, 4.7.1, 4.7.2	(M105)
Initiator	Locked shut requirements	3.8, 4.7.3	(C1)
Initiator	Packaging, packing and container markings	section 5, 4.5.1	(M106)
Initiator	Outer sleeve seating torque verification	3.6, 4.6.4	(M107)

4.5.1 Packaging, packing and marking. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in section 5.

4.6 Tests.

4.6.1 Cartridge verification. The contractor shall provide verification of cartridge conformity with the requirements of 3.3.1.

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4.6.2 Visual inspection. Examine all initiators and reject those having any of the defects of 3.4.

4.6.3 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.5 shall be ascertained by means of a test circuit which limits the bridge current to 25 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate within one percent in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.5 shall be rejected.

4.6.4 Torque test. Prior to temperature conditioning and firing, each of the sample initiators of 4.3.2, 4.7.1 and 4.7.3 shall have a seating torque of 300 $\pm$ 10 inch pounds applied to the initiator outer sleeve. Failure of any initiator to meet the requirements of 3.6 shall cause rejection of the entire production lot or preproduction sample.

4.7 Ballistic and locked shut tests.

4.7.1 Sampling. A sample of initiators shall be selected at random from each production lot for the ballistic test of 4.7.2 in accordance with the requirements of the single sampling plan of MIL-STD-105, inspection level 1, and an acceptable quality level (AQL) of 0.40. In addition to the sample of initiators for the ballistic test, 30 additional initiators shall be selected at random from each production lot. Of the 30 initiators, 20 shall be fired in the locked shut test of 4.7.3 and 10 shall be retained for investigative purposes.

4.7.2 Ballistic test. The sample of initiators shall be conditioned for a minimum of 6 hours at $70^{\circ} \pm 5^{\circ}$ F and fired in the closed chamber test fixture DL 298AS116. The firing current shall be 5 $\pm$ 1.0 - 0.00 amperes direct current.

4.7.2.1 Criteria of acceptability. Any initiator which fails to meet all requirements of 3.7.1, 3.7.2, and 3.7.3 shall be considered a defective unit. Lot acceptability shall be in accordance with table II-A of MIL-STD-105 using an AQL of 0.40.

4.7.3 Locked shut test. A sample of 20 initiators from each production lot shall be conditioned for a minimum of 6 hours at $70^{\circ} \pm 5^{\circ}$ F

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and shall be fired in a fixture that prevents the gases from venting through the initiator Firing current shall be 5 + 1.0 - 0.0 amperes direct current.

4.7.3.1 Criteria of acceptability. Any initiator of the production lot samples (ballistic or locked shut) that exhibits mechanical failure of any metal or plastic initiator component which results in failure to contain the propellant gases shall cause rejection of the lot represented. After firing, the ignition element end of each CCU-23/B cartridge of the initiator production lot samples shall be examined for evidence of propellant gas leakage through or around the ignition element. If one or more cartridges of the initiator production lot samples exhibit evidence of propellant gas leakage the production lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four initiators shall be packaged in an inner container in accordance with Drawing 275AS108.

5.1.2 Level C. When directed by the procuring activity, those initiators which are intended for expenditure in preproduction sample and production lot acceptance tests shall be packaged in accordance with method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 171-178.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Twelve inner containers as described in 5.1.1 shall be packed in accordance with Drawing 275AS107.

5.2.2 Level C. When directed by the procuring activity, initiators intended for expenditure in preproduction sample and production lot acceptance tests, packaged in accordance with 5.1.2, shall be packed to afford protection against damage during direct shipment from the

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supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on Drawings 275AS108 and 275AS107, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 Intended use. The JAU-1A/B cartridge actuated initiator is designed to eject search stores from antisubmarine warfare aircraft.

6.2 Ordering information. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether preproduction inspection is required and, if so, specify the test activity (see 3.1)

c. Production lot size and test activity (see 4.3.3)

d. Items of data required for each preproduction and production lot (see 6.3)

e. Levels of packaging and packing (see 5.1 and 5.2)

f. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this purchase description is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition Ashore," OP 5, are applicable.

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6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

6.3.1 Test report. The contract or order should specify that results of ballistic tests are to be forwarded to the Naval Air Systems Command (AIR-53222A) and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity, and destination of data cards.

6.4 Definitions. For the purpose of this specification, the following definitions apply.

6.4.1 Packaging.

6.4.1.1 Level A. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration, and physical damage during handling, shipment, indeterminate surface, and world-wide redistribution.

6.4.1.2 Level C. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Packing.

6.4.2.1 Level A. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.2 Level C. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general

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will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

Preparing activity:

NAVY (AS)

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